Group Topic: Big Data Analytics for Student Dropout Prediction in Higher Education

Git Repo/ Colab Link: https://github.com/prinxe0/Big-Data-Analytics-for-Student-Dropout-Prediction-in-Higher-Education

Lab No.: Deliverable 4

Date: May 16, 2026

Objectives

The objective of this activity is to design, generate, and integrate comprehensive data visualizations that communicate the findings of the exploratory data analysis and predictive machine learning models. Specifically, the task aims to translate complex big data metrics and algorithm outputs into interpretable, executive-ready visual summaries that clearly illustrate the relationships between student mental health factors and dropout risk.

Tasks Completed

1. Data Visualization and Insights Generation
   a) Generated a comprehensive Feature Correlation Heatmap using Seaborn to illustrate the compounding effects of psychological variables (e.g., the strong positive correlation between stress, anxiety, and burnout).

   b) Created the Risk Level Percentage Distribution chart and Risk Score Distribution histogram to visually communicate the dataset's class imbalance and population thresholds (highlighting the critical 1.5% high-risk segment).

   c) Designed a Feature Importance bar chart to explicitly map the mathematical impact of both Risk Factors (positive weights) and Protective Factors (negative weights) assigned by the scoring model.

   d) Rendered a high-resolution Confusion Matrix heatmap to visually validate the classification model's accuracy and the density of true-positive predictions across the Low, Medium, and High-risk tiers.

Tasks to be Completed

1. **Interactive Dashboard Integration:** Transition the static visualizations into a dynamic, interactive dashboard framework (such as Streamlit, Tableau, or PowerBI) to allow stakeholders to explore the dropout risk factors during the final capstone presentation.

2. **Final Manuscript Compilation:** Finalize the comprehensive written project manuscript, seamlessly embedding these visual artifacts into the Results and Analysis sections to support the project's data-driven conclusions.

Results and Analysis

The visualization phase successfully translated the abstract data engineering pipeline and machine learning outputs into clear, actionable insights. By graphically representing the feature importance and risk score distributions, it became visually apparent that severe psychological exhaustion acts as the primary catalyst for dropout risk, heavily outweighing standard academic pressure. Conversely, the visualizations confirmed that a strong mental health index and social support serve as the absolute strongest protective factors. These visual artifacts effectively validate the model's predictive capabilities and provide a clear, data-backed narrative for identifying at-risk student populations.

Challenges and Solutions

Challenge: Representing the massive scale of the 1,000,000-row dataset in statistical plots without causing severe rendering lag, local memory crashes, or creating visually incomprehensible, over-plotted charts.

Solution: Avoided plotting individual data points directly. Instead, utilized Apache Spark's distributed aggregation functions (such as .reduceByKey() and .collectAsMap()) to compute the summarized statistics on the cluster. Only the minimized, aggregated summary data (e.g., bin counts, matrix totals, and percentage distributions) was brought to the local driver node for rendering via Matplotlib and Seaborn, ensuring fast and stable visualization generation.

Conclusion

The core objective for this deliverable was successfully met. By designing and generating professional-grade statistical visualizations, the complex and highly dimensional relationships within the large-scale Student Mental Health and Burnout dataset were made highly interpretable. These visual assets successfully bridge the gap between backend big data processing and stakeholder decision-making, fully preparing the project for its final presentation and practical deployment.